Leveraged Buyouts and Private Equity

Week 9: Mergers, MBOs & other Corporate Reorganizations

10 March 2026

Professor Julian Franks

I am grateful to Faisal Ramzan, Partner in Proskauer Rose and Tim Jenkinson for advice and data.

Outline

1. What is special about Private Equity?
2. Structure of a PE Transaction & Pricing
3. How profitable is private equity
4. A study on sources of profitability

1. What is special about PE?

- PE is an alternative to trade sale, a spin-off or carve out.
- New vehicle funded by 1/3 equity from investors, *limited partners* (LP), & 2/3 debt from banks & special funds.
- Vehicle invests in portfolio companies managed by a *general partner* GP. GP arranges financing and appoints board of directors. GP like a fund manager.
- Use of high yield debt ('junk debt') or mezzanine finance (sits in the middle of the financing structure), can include payment in kind (PIK), warrants etc.
- Mean return to mezzanine (1984-2019) 11.7% (Preqin)
- High leverage motivated by tax benefits, reduced agency costs/improved incentives & potentially low cost of debt.

Incentives

- One impact of high leverage is to reduce the free-cash flow available to management
  - Jensen ('The Eclipse of the Public Corporation') regards PE as an antidote to governance failures, & reduce the agency costs of dispersed ownership.
- High leverage increases risk of financial failure – forces management to profit maximize. Increase incentives by remunerating them with equity stake
- Private Equity uses financial engineering to reduce costs of high leverage.
- In taking private BAA (Heathrow-Gatwick airports) Ferrovial used securitised vehicle with
- > 80% leverage. Securitised transaction. Debt secured by regulated revenues.

Tax shields

- Leverage creates tax shields at the corporate level, reduces cost of capital
- Knauer et al (2014) estimated tax shields from leverage for 56 German LBOs contributed 20% of the enterprise value.
- Jenkinson & Stucke (2011) estimated incremental tax shields in US worth 8-11% of enterprise value & 65-87% of deal premium
- Capital gains on sales may be taxed at a low rate: 10% in UK
- Some countries have imposed limits on tax-deductibility of debt (e.g. Germany)

Source: How much do private equity funds benefit from debt-related tax shields? Alexander Knauer, Alexander Lahmann, Magnus Pflücke, and Bernhard Schwetzler, Journal of Applied Corporate Finance, March 2014.

High-Leverage PE Transactions, Case Study:

Blackstone's Acquisition of Emerson's Climate Technologies (2022)

- Blackstone acquired a **55% stake in Emerson's Climate Technologies** in a deal valued at **$14 billion**.
- The deal was **highly leveraged**, using structured debt financing to optimize capital efficiency.
- Blackstone used **securitized debt** to finance a significant portion of the transaction.

- Blackstone in 2026 manager of $416 billions in PE assets.

Mispricing of debt and risk?

- Mispricing suggests an obvious arbitrage opportunity – is that realistic?
- One case of 'perceived' market inefficiency was the credit boom in early 2000s– until Summer 2007 – which had all the hallmarks of a bubble
  - Lending multiples hit all-time highs
  - Interest rates, if anything, fell despite the increase in lending multiples
  - Covenant-lite loans
  - Citibank underwrote loan for £4.2 billions buyout of EMI by PE house Terra Firma (lost $2 billions on loan upon default)

Case Study: KKR's Acquisition of FiberCop (2023)

- In 2023, KKR acquired FiberCop, an Italian telecommunications company, as part of a €22 billion investment strategy. ft.com
- Post-acquisition, FiberCop faced significant challenges, including the resignation of its CEO due to conflicts with KKR and a substantial downward revision of earnings forecasts of $449 Milions).
- These issues have raised concerns about potential dividend cuts or the necessity to increase debt, highlighting the risks associated with aggressive debt financing and over-optimistic projections.

2. Structure of a deal and pricing

A Typical Buyout: Purchase of Kwik-Fit by PAI Partners in August 2005

	Amount (£m)	Terms	Pricing (spread over LIBOR)	Multiple of EBITDA (EBITDA≈£95m)
Enterprise Value	**773.5**			**8.1 x**
Equity	**191.0 (25%)**			**2.0 x**
Debt				
Term Loan A	140	7 year amortizing	2.25%	
Term Loan B	135	8 year bullet	2.50%	
Term Loan C	135	9 year bullet	3.00%	
Total Senior Debt	*410.0*			*4.3 x*
2nd Lien	75	9.5 year	5.00%	
Mezzanine	97.5	10 year	4.5% + 5% PIK	(pay-in-kind)
Total Debt	**582.5 (75%)**			**6.1 x**
Revolving credit facility	40	7 year	2.25%	
Capex facility	50	7 year	2.25%	

Source: Axelson, Jenkinson, Stromberg and Weisbach, 2008.

Multiples

- In LBO transactions, valuations (enterprise values) and leverage are typically expressed in terms of multiples of
  - Normalized EBITDA (last 12 months, or pro-forma, or forecast)
  - EBITDA – (maintenance?) CAPEX
  - EBIT (LTM LAST 12 months)
- Although traditional discount cash flow analysis is used, the buy-to-sell nature of PE deals increases the focus on market valuations

$$\text{EV/EBITDA multiple} = \frac{\text{EV}}{\text{LTM EBITDA}}$$

$$\text{EV/EBITDA - Capex multiple} = \frac{\text{EV}}{\text{LTM EBITDA - Capex}}$$

$$\text{EV/EBIT multiple} = \frac{\text{EV}}{\text{LTM EBIT}}$$

Illustrative Private Equity Transaction

- Private company: ACME Inc.
  - Company's net debt: $400
  - Selling shareholders want $600 for the company
  - Company's LTM EBITDA: $125 => EV/EBITDA multiple: 8.0x
  - D&A/Capex margin: 10%
  - Tax rate: 35%
- Forecast
  - Intended holding period: 6 years
  - EBITDA CAGR: 3.1%
  - EV/EBITDA multiple after 6 years: 10x
- Financing
  - Senior debt: 3.2x EBITDA ($400) at 4.4% interest (amortised out of annual net earnings)
  - Junior debt: 2.4x EBITDA ($300) at 8.0% interest (Bullet repayment)

Illustrative LBO: cash flows to enterprise & equity

		1	2	3	4	5	6
Senior Debt at End of Year	400.0	351.5	299.4	243.3	183.2	118.8	50.0
Junior Debt at End of Year	300.0	300.0	300.0	300.0	300.0	300.0	300.0
EBITDA	125.0	128.9	132.8	136.9	141.2	145.5	150.0
D&A / Capex		12.9	13.3	13.7	14.1	14.6	15.0
EBIT		116.0	119.6	123.3	127.1	131.0	135.0
Interest on Senior Debt		17.4	15.3	13.0	10.6	8.0	5.2
Interest on Junior Debt		24.0	24.0	24.0	24.0	24.0	24.0
EBT		74.6	80.3	86.2	92.5	99.0	105.9
Taxes		26.1	28.1	30.2	32.4	34.7	37.1
Net Earnings		48.5	52.2	56.1	60.1	64.4	68.8
Enterprise Value	1,000						1,500
Equity Value	300						1,150

Illustrative LBO: cash flows to enterprise & equity

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Sr debt outstanding	400.00	351.65	299.58	243.61	183.55	119.21	50.40
Jr debt outstanding	300.00	300.00	300.00	300.00	300.00	300.00	300.00
EV/EBITDA multiple	8.00						10.00
EBITDA	125.00	128.88	132.87	136.99	141.24	145.61	150.13
D&A		12.89	13.29	13.70	14.12	14.56	15.01
EBIT		115.99	119.58	123.29	127.11	131.05	135.12
Sr debt interest		17.60	15.47	13.18	10.72	8.08	5.25
Jr debt interest		24.00	24.00	24.00	24.00	24.00	24.00
EBT		74.39	80.11	86.11	92.39	98.98	105.87
Taxes		26.04	28.04	30.14	32.34	34.64	37.05
After-tax earnings		48.35	52.07	55.97	60.06	64.33	68.82
Enterprise value	1,000.00						1,501.28
Debt	700.00						350.40
Equity	300.00						1,150.88

Assessing profitability

- Compare Enterprise IRR with WACC
- Private Equity firms often compare IRR on Equity Cash Flows with (levered) cost of equity.
- Problems with these comparisons?

  Leverage is changing over time so WACC/Levered cost of equity is also changing over time. Complicated to calculate the annual after tax discount rates.
- One solution: use Adjusted Present Value Approach (APV)
- See Ramin's slides in Session 1 Mergers FO1B and his Review of Valuation Methods. See also Appendix 1 for an example of APV calculation.

APV calculations

- Step 1: Value unlevered investment (Unlevered CF (Yr1)=115.99*(1-0.35)=75.39

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Unlevered cash flow	-1,000	75.39	77.73	80.14	82.62	85.18	1,589.11
Return on assets	10%						
Unlevered EV	199.32						

Step 2: Value tax shield

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Interest payments		41.60	39.47	37.18	34.72	32.08	29.25
Interest tax shield		14.56	13.82	13.01	12.15	11.23	10.24
Cost of debt		6.06%	6.20%	6.39%	6.63%	6.98%	7.48%
Discounted tax shield		13.73	12.25	10.81	9.40	8.01	6.64
PV tax shield	60.84						

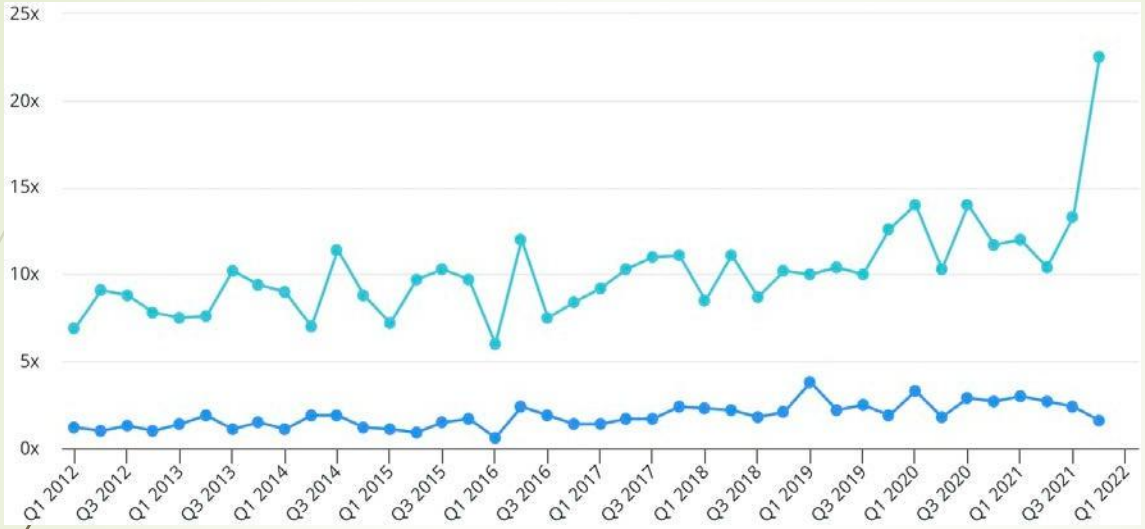
APV = 260.16

Why might APV give a different value to WACC

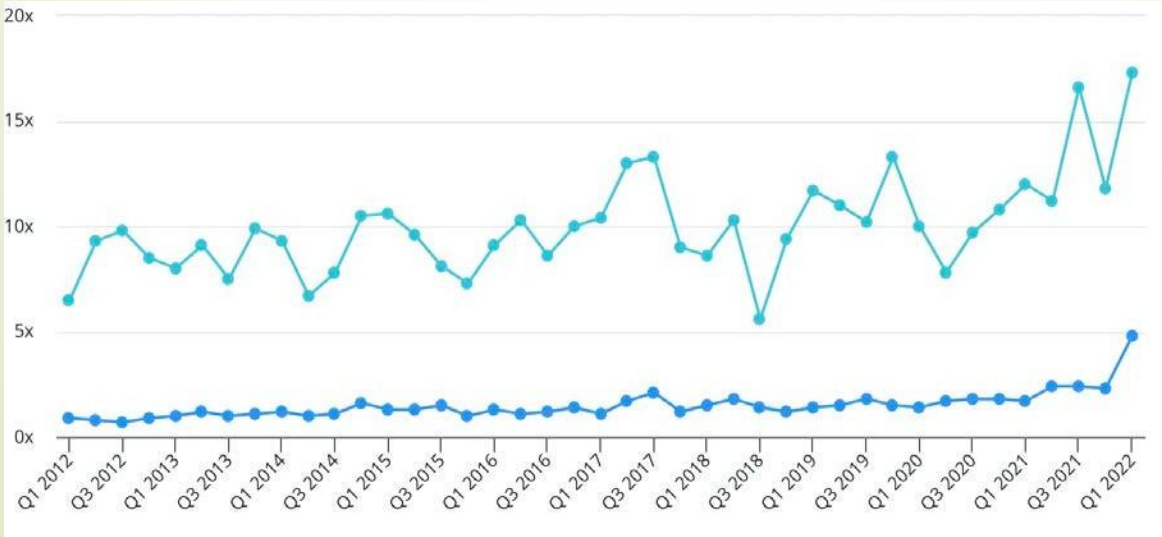
- Constant WACC that does not reflect accurately changing leverage over time
- Tax shields in APV discounted at cost of debt. Textbook WACC assumes tax shields discounted at unlevered cost of capital
- When should you use cost of debt to discount tax shields in APV and when should you use unlevered cost of capital?
- Use cost of debt when tax shields are low risk (e.g. leverage is fixed in dollar terms).
- Use unlevered cost of capital when debt is a constant proportion of market value (or present) value of firm. In the latter case, assumption is you relever if enterprise value increases and delever if enterprise value goes down. Thus, riskiness of tax shields is similar to the risk of underlying assets.

PE-backed Buyout Deal Multiples, Q12012-YTD2022

US



Europe



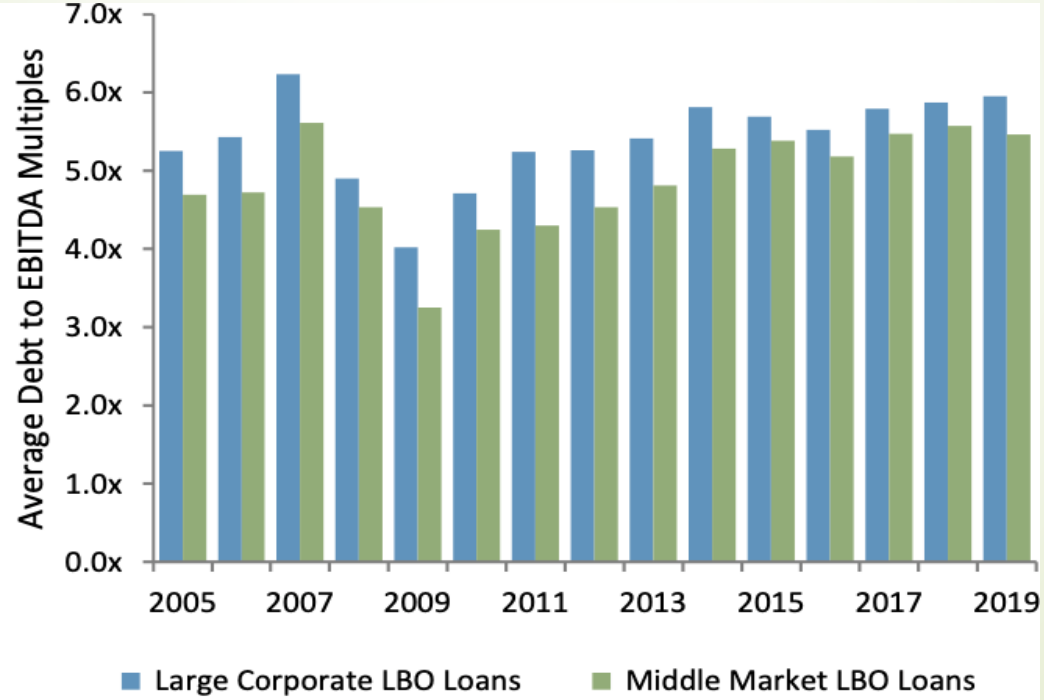
Median EV/EBITDA

Median EV/Revenue

Source: Preqin

Total debt to EBITDA ratios

The metric most widely used to measure risk in a debt financing transaction is the level of debt to EBITDA.



Source: S&P LCD, Quarterly Q4-19 Leveraged Lending Review.

Deal sizes over time

- 2017 there were no deals > $10 bln
- 2018 largest deal $21 billion (Dr Pepper Snapple)
- 2023 $13 billion bid to take Toshiba private by Japanese investors
- 2024: $58 billion management buyout proposal for Seven & i Holdings. Did not go ahead.
- In 2025 largest deal on record $55 billion Electronic Art taken private by a syndicate of firms!

State of the market in recent years

- Rise of large debt funds e.g. Ares, Alcentra. Important in mid market.
- Unitranche loans: hybrid loan package and sourced from one supplier, blended interest rate, lower transactions costs
- EBITDA multiples 5 to 5 ½ times even 6 times
- Interest rates LIBOR + 6%
- PIK finance 9-11%
- Compare this to Kwik Fit Deal
- How has it changed in 2026?

FT March 6, 2025: 'Private equity industry's assets shrink', struggled to sell assets. But in 2025 highest deals on record >$1 trillion. Message considerable volatility!

Feb 2026: FT 'Private Equity Backlog hits all time high of $4 trillion in unsold investments.

Profitability of Private Equity for

a

$14 billion dollar endowment

fund?

During the year that ended June 30, 2021, investment markets generously rewarded all the central hallmarks of our investment strategy: active management, equity risk, growth, innovation, and a long time horizon. We're very pleased for our shareholders by the unprecedented return of 49% but recognize that this is a single, highly unusual point in time. I am confident that UVIMCO and the University's combined strength will help us steer the course ahead. We will seize the opportunities presented and navigate the risks together to ensure the continued prosperity of the University for future generations to come."

— Robert Durden, CFA
Chief Executive Officer/Chief Investment Officer

TABLE 1 UVIMCO Long Term Pool Strategy Allocation and Investment Returns

In percentages, as of June 30, 2021

			Annualized			
	Allocation [1]	1YR	3YR	5YR	10YR	20YR
Long Term Pool [2]	100.0	49.0	18.4	15.8	12.1	10.6
Policy Benchmark [3]		29.4	13.0	11.6	8.8	7.4
Public Equity	29.9	51.4	19.4	19.5	14.3	13.6
Long/Short Equity	17.0	33.3	14.1	11.5	9.7	9.3
Private Equity	26.4	98.7	40.6	29.3	21.8	14.7
Real Assets	10.0	49.0	5.8	10.6	9.3	9.9
Marketable Alternatives	5.2	4.8	0.9	3.0	4.5	4.0
Credit	4.8	29.9	9.4	9.0	8.0	9.1
Fixed Income	3.9	1.4	7.6	4.5	2.5	4.9
Cash	2.8	0.0	1.2	1.0	0.5	-

Where have the returns from PE come from? A UK study

Based upon a paper and presentation by Viral Acharya and Conor Kehoe (Head of Private Equity at McKinsey)

Key questions addressed as part of the research

1

Do Private Equity owned companies create additional value at an enterprise level vs. their quoted sector peers?

1a

Are the returns to large, mature PE houses simply due to financial gearing of deals on top of comparable sector risk?

1b

What is the nature of systematic and leverage-induced risk in PE deals?

2

What is the operating performance of PE owned companies relative to that of quoted
peers? How does this performance relate to value creation?

3

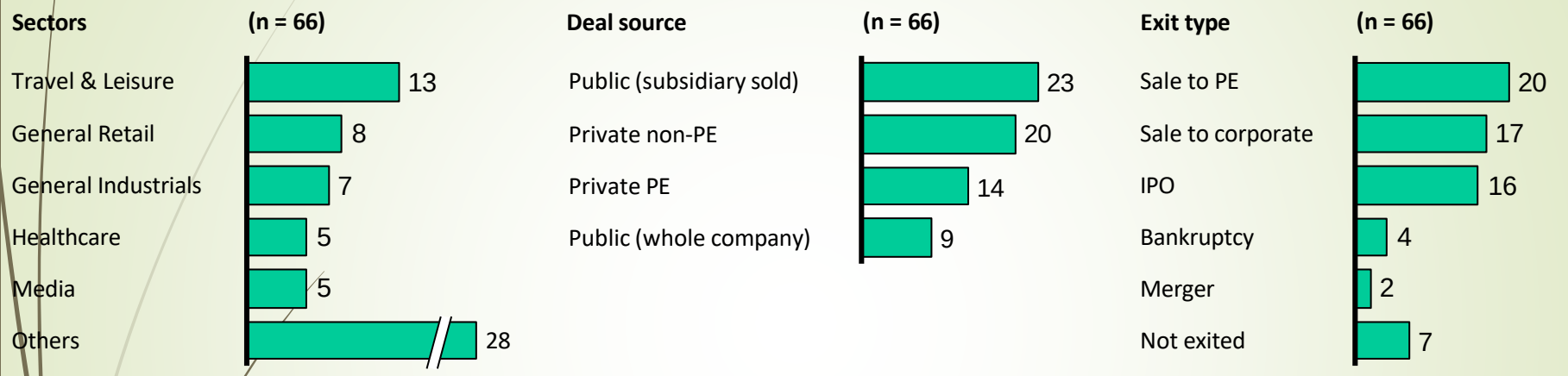
What are the distinguishing characteristics of the PE governance approach relative to PLC Boards? Which of these are associated with value creation?

Focus on UK deals: characteristics of data-set

- Deals for UK companies acquired by 12 large and mature PE houses. McKinsey – Acquired between 1996–2004, (if) exited during 2000-2007
- Deals have a median enterprise value at start of ~ €471 million
  - 18 deals > €1 billion
  - 15 in €500m- €1billion
  - 33 in €100m- €500m
- Currently, the data-set comprises 66 UK deals:
  - 59 exited deals (4 bankruptcies)
  - 7 non-exited deals for which end EV cash flow simulated assuming start EV / EBITDA multiple and applying to end year EBITDA
- 29 have acquisitions and/or divestments; the balance 37 are "organic" deals
- Deals were held for an average of 3.8 years by PE firms

Distribution of deals by sector, deal source, exit type and years

Deals split by…



Deals by entry and exit year (n = 66)

Years	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
Entry	2	5	7	9	8	6	10	12	7	n/a	n/a	n/a
Exit	n/a	n/a	n/a	n/a	2	4	6	6	10	14	19*	5**

* Includes five deals for which exit simulated

** Includes two deals for which exit simulated

Interpreting the returns

- Benchmark returns 11%
- Abnormal returns 18.3%
- 11% come from leverage, 7% is alpha
- Crucial organic deals perform best (see the next slide).
- Note returns include fees to general partners. Lower returns to limited partners?

Organic deals outperform inorganic deals on both total IRR and alpha; divestments appear to distinctly under-perform in our sample

No. deals	Deal Type	IRR	Median / StdDev	Cash multiple	Alpha	Median / StdDev	Cash multiple
37	Organic*	47.1	40.3 / 48.8	2.9	15.3	14.3 / 25.4	0.9
Inorganic 16	Deals w/acquisitions	30.0	33.5 / 34.2	3.3	3.8	2.0 / 17.8	-0.1
Inorganic 13	Divestments	9.7	17.4 / 34.7	1.9	-2.7	-1.6 / 16.5	-0.1
66	Total	35.6	31.0 / 45.0	2.8	9.0	9.2 / 23.2	0.5

	Deal Type	IRR	Median / StdDev	Cash multiple	Alpha	Median / StdDev	Cash multiple
33	Organic*	36.6	37.5 / 38.1	2.5	13.0	13.0 / 25.4	0.7
Inorganic 16	Deals w/acquisitions	30.0	33.5 / 34.2	3.3	3.8	2.0 / 17.8	-0.1
Inorganic 13	Divestments	9.7	17.4 / 34.7	1.9	-2.7	-1.6 / 16.5	-0.1
62	Total	29.3	28.2 / 37.4	2.6	7.3	8.8 / 22.6	0.3

* Organic deals are those where no major acquisition or divestment was reported

Source: PE deal data; Datastream; team analysis

Company outperformance driven by active ownership practices deployed mainly before or right after acquisition

Pre

Post

Sourcing

Due diligence

Deal structuring

First 100 days

Medium/ long term

Exit

1. Proprietary Deal insight
- Leverage privileged knowledge
- Bring board members and expertise

2. Value creation plan
- Create a new value creation plan
- Refine plan during first 100 days
- Adjust/ implement systems to monitor plan evolution
- Immediately react to plan deviations

3. Early manage-ment changes
- Strengthen manage-ment before closing
- Find addi-tional repla-cements within first 100 days

4. Substantial but focused incentives
- Very significant value at stake for top executives

5. Significant time spent upfront
- Invest significant time in first 100 days (>1,5 days per week)
- Interact with CEO/CFO multiple times a week

6. External support
- Use external support both in the first 100 days and in the medium/ long term

Observations on active governance in PE model

- High level of senior management turnover
- Plan changes significantly in a high number of cases/active external support to the board.
- Smaller boards compare with listed company
- PE partners much more involved more meetings (weekly) and longer
- Deals with large alpha depend upon growth in margins.

Case Study: KKR's Investment in PetVet Care Centers (2018–2024)

- KKR acquired PetVet Care Centers in 2018, focusing on scaling the veterinary clinic chain through acquisitions and operational improvements.
- Following the PE governance model, KKR replaced top management within 100 days, centralized procurement, and improved efficiency.
- The firm's EBITDA margins expanded, driving up valuation.
- In 2024, KKR exited the investment by selling PetVet to another PE firm at a much higher valuation.

Summary

1. PE deals of large, mature houses in the UK have out-performed their quoted sector peers on average, based on "alpha", controlling for sector risk and leverage
2. Harris, Jenkinson & Kaplan (2015) measure returns to **Limited Partners** on 781 private equity funds over 1984-2005: they earn 3% per year excess returns over public equity benchmark.
3. Out-performance correlated with stronger operating performance relative to peers; Improving margins seems mantra for generating alpha (organic deals)
4. Alpha deals are associated with replacing management with a new team focused on improving margins and profitability per employee and provision of support to top management from GPs and external experts
5. Less outperformance in recent years (see 'Thirty years after Jensen prediction, is Private Equity a superior form of ownership?'

Appendix: Project valuation

You only need to understand the APV method for this course. The rest of the material in this appendix is optional

Professor Julian Franks

Calculating discount rates & PV for project financing transactions e.g. PE, infrastructure

- In a PE or infrastructure transaction the debt to value ratio depends upon the (present) value of the project's cash flows. That debt to value ratio is changing over time since free cash flow is often used to repay debt. The result is that if you use the WACC you must use different discount rates for different periods.
- If you know market value or present value of the assets every period then it is easy to calculate the market value leverage ratio and the discount rate.
- But suppose you don't know the value of the assets, because that is what you are trying to find out. Then it is more complex to calculate the leverage ratio and the discount rate.
- In this case you must solve for the leverage ratio and the value of the assets simultaneously.
- Is there a simple approach to this problem?

Valuation of the project

Value a PE firm with the following constant expected annual CFs in perpetuity:

EBIT	600	
Interest payments		60
Taxable income		540
Taxes (20%)	108	
After tax income		432

The asset beta is 1.2. The risk free rate is 7%, the expected return on (high yield) debt is 10%. Assume the debt beta is 0.5, and the risk premium on the market is 6%. The tax rate is 20%.

The firm has debt of £600 outstanding in perpetuity. The book value of the assets is currently £3000.

Which FCF do we use in valuing the enterprise value with a WACC or APV calculation? Is it

432, 492 or 480? See next page for the answer

(i) **Calculate the Adjusted Present Value (APV) of the firm.**

APV = Value of all equity financed firm + PV of Financing Benefit

Step 1: calculate the all equity i.e. the unlevered discount rate

R_a = 7% + 1.2 (6%) = 14.2%

PV of all equity financed firm:

All equity cash flows 600 (1-.2) = 480

PV = 480/0.142 = **3380.28**

Step 2: calculate PV of tax shield on leverage = (60 x .2)/0.142 = **84.51**

PV (firm) = 3380.28 + 84.51 = **3464.79**

1. The all equity cash flows are cash flows before interest but after taxes. The tax calculation omits the interest tax shields since these are separately valued in Step 2. Don't double count!
2. Tax shields are assumed as risky as the operating cash flows. This is assumption of our WACC formula.

(ii) If we c**alculate a WACC we must know the market leverage ratio, how do we do it?**

Textbook WACC

What is the market value gearing ratio. Problem: you do not know the value of the firm i.e. the enterprise value. You could take the easy way out and use the gearing ratio (600/3464.79) from the APV answer. But you may have to show the WACC calculations. We do it by iteration. First, our initial estimate of the gearing ratio is based on the book value of the assets i.e. D/V = 600/3000 = 0.2

WACC = i (1-T) D/V + R_e E/V

WACC = .1(1-.2) .2 + R_e x .8

Where R_e = 7% + 1.375 (6%)

= 15.25%

$\beta_a = \beta_e$ E/V + β_d D/V

1.2 = β_e x 0.8 + 0.5 x 0.2

β_e = 1.375

WACC = .1(1-.2) .2 + .1525 x .8

= 13.8%

PV = 480/.138 = **3478.26**

This is not our final value of the project because the gearing ratio based on our new estimate of the value of the firm: 600/3478.26 = .1725 is not the same as our initial one of 20%. Our initial debt ration of 20% must equal our final debt ratio. We have not converged.

Step 1: Recalculate a new β_e based on leverage estimate of 17.25%

Use formula for beta asset: β_a = β_e E/V + β_d D/V

1.2 = β_e x 0.8275 + (.5 x .1725)

β_e = 1.3459

Step 2: Recalculate a new R_e = 7% + 1.3459 (6%) = 15.076%

Step 3: Calculate WACC = .10(1-.2) .1725 + .15076 x .8275 = .13855 or 13.855%

PV = 480/ .13855 = **3464.45**

We could continue iterating using a new gearing ratio of 600/3464.45 or 17.3%. We can see this is close to our starting debt ration of 17.25%. Or we can just compare it with the APV value.

Notice that could have used the gearing ratio from the APV (600/3464.79) = .1731 and then calculate the WACC.

An important assumption in these WACC calculations: the firm's tax shields are risky

- APV & WACC gave the same answers because we assumed that the tax shields were risky and were discounted at R_a (and not at R_d). Risky because leverage ratio is constant and therefore dollar leverage varies with profitability and value. This is the assumption imbedded in the particular WACC formula we have used. Thus, if value of firm doubles (halves) the $ leverage must double (halve) to keep the leverage ratio constant.
- But suppose debt is fixed & constant in dollars. The tax shields are then less risky than the asset's cash flows & we should use a lower discount rate. What rate,? say R_d. then the two methods will not give the same answer.

What should you do?

If leverage ratio is not constant and debt is fixed in $ terms textbook WACC is not correct: 2 solutions

- Use APV and discount tax shields at R_d.

- Or, use another WACC method: MM WACC

  WACC (mm)= r (1$-T_c$D/V) where r is return on assets

- But here is a catch. For MM WACC there is a different formula for levering/unlevering beta:

  β_e = β_a + (1-Tc)(β_a − β_d)D/E

See Brealey Myers and Allen, 13 th edition: pages 522-523

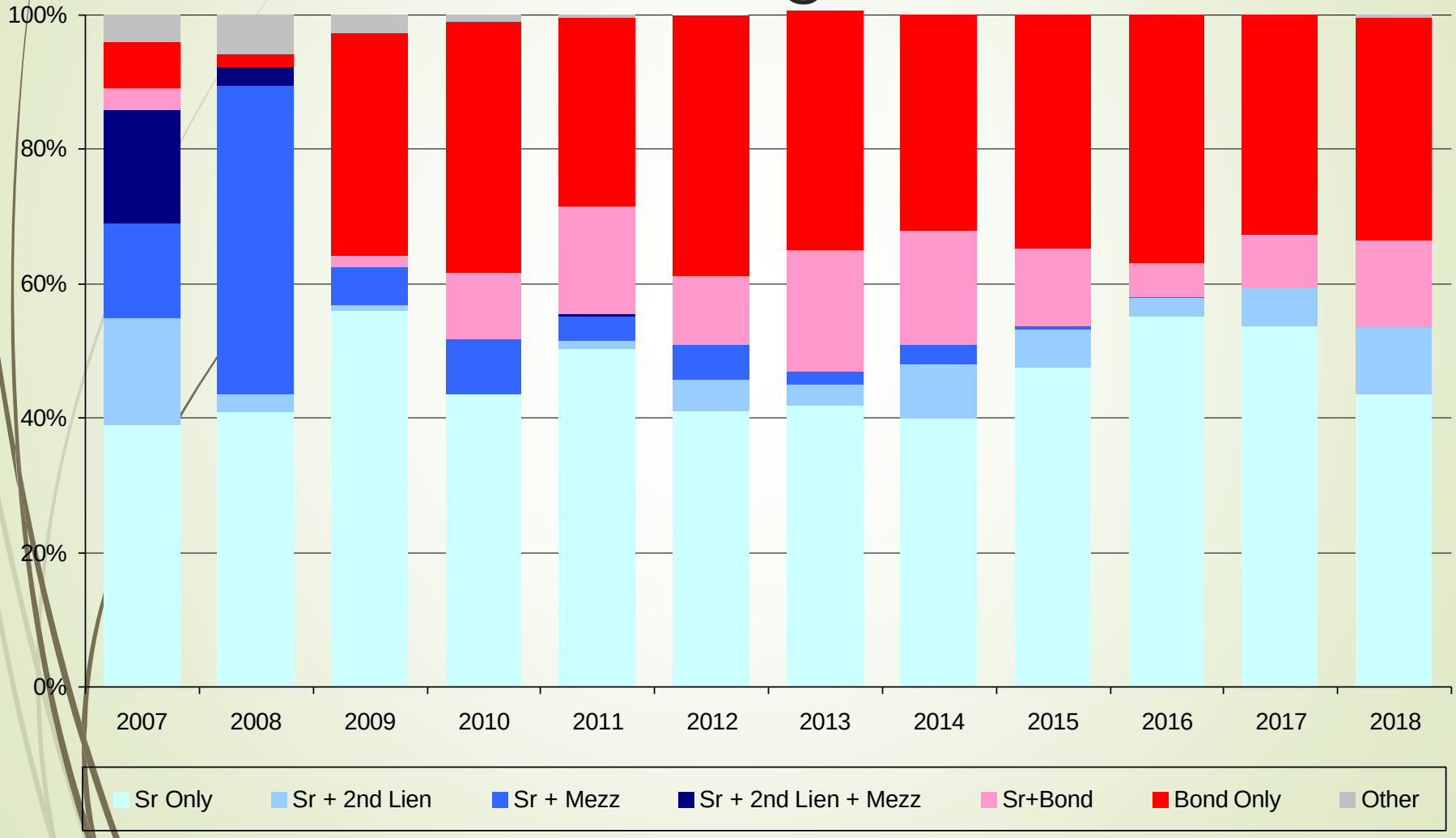
Appendix 2

Seniority classes of debt

Senior secured debt (first lien)
Senior secured debt (second lien)
loans

Senior unsecured **debt**
Senior subordinated debt
Subordinated debt
Junior subordinated debt
(mezzanine loans)
bonds
(bridge loans)

mezzanine
and high-yield

recovery rate at default

European borrowers' sources of funding